3-State EV Infrastructure Gap Analysis

A network-distance coverage gap analysis applied independently across Tennessee, Oklahoma, and Mississippi's designated EV charging corridors, identifying exactly where federal spacing requirements aren't yet met

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The Problem

Federal infrastructure funding decisions under the NEVI program hinge on one core spatial question for each designated corridor: where along that corridor is there currently no compliant fast charging within the required spacing, and how much of the state's corridor mileage does that represent. Answering that question rigorously, and consistently, across three separate state DOT clients required more than eyeballing a map of existing chargers.

- Existing charger data doesn't distinguish, out of the box, between a station that actually meets the federal compliance bar (a minimum port count and simultaneous power output) and one that doesn't, so a raw charger location dataset overstates real coverage if used naively.
- Corridor mileage isn't a single number. Each state's designated corridor system is a network of separate named routes, and coverage or a gap on one route doesn't say anything about coverage on another.
- The 50-mile maximum spacing rule is a network distance constraint along the corridor itself, not a straight-line distance from any point, so a defensible gap analysis has to measure distance the way a vehicle actually travels the corridor, not as a radius.
- Three states means three different existing charger baselines, three different corridor lengths and route structures, and three different funding cycles, so the analysis needed to produce results that were comparable in method even though the underlying geography and inputs differed completely.
- A gap number by itself isn't actionable without knowing which specific stretch it refers to, so the analysis needed to tie every identified gap back to an actual location a planner could act on, not just a statewide percentage.

My Approach

I built a coverage gap analysis applied independently across Tennessee, Oklahoma, and Mississippi's designated Alternative Fuel Corridor systems, using the same measurement method in all three so the results were comparable in how they were derived, even though each state's corridors, existing chargers, and funding structure were entirely different.

A compliance filter before anything else. Before any gap could be measured, every existing charger record was checked against the federal technical bar, a minimum number of ports capable of a minimum simultaneous power output, so only chargers that actually satisfy the funding program's requirement count as covering a stretch of corridor. A charger that exists but doesn't meet that bar is treated as if the gap it sits in is still open.

Gaps measured the way a corridor is actually driven. Distance between compliant chargers is measured along the corridor network itself, not as a straight line, so a 50-mile gap in the analysis reflects 50 miles a driver would actually need to cover without a compliant charging option, consistent with how the federal spacing rule is written.

Every corridor treated as its own coverage segment. Rather than rolling every designated route into one statewide number, each corridor, each interstate or highway carrying the alternative fuel designation, gets its own gap analysis, so a planner can see exactly which route has the worst remaining coverage rather than a single blended statistic that hides where the actual problem is.

One method, three independent baselines. The same compliance filter and network distance measurement ran against Tennessee's, Oklahoma's, and Mississippi's own corridor networks and charger inventories separately, so each state's gap analysis reflects its own real infrastructure, while still being directly comparable to the other two in terms of how the numbers were derived.

Gaps tied to a specific, actionable location. Every identified gap is anchored to the actual stretch of corridor it falls on, not reported as an abstract statewide percentage, so the output functions as a prioritization map, not just a summary statistic, feeding directly into where each state's next funding round should focus.

Technical Highlights

- **Data sources:** State DOT and FHWA corridor designations, DOE Alternative Fuels Data Center existing charger data, state program compliance verification records
- **Method:** Federal compliance filtering applied to raw charger inventories before any spatial analysis, network distance measurement along each corridor rather than Euclidean buffering
- **Scope:** Independent gap analysis across Tennessee, Oklahoma, and Mississippi's full designated Alternative Fuel Corridor networks
- **Output:** Per-corridor gap segments identifying exactly where the 50-mile compliant-charger spacing requirement is not currently met
- **Consistency:** Identical measurement methodology applied to three states with entirely different corridor structures and charger baselines, producing comparable results despite different underlying data

Why It Matters

A gap analysis is only useful if it tells a planner exactly where to focus, and it's only trustworthy if the underlying distance and compliance logic actually matches the rule the funding program is enforcing. Filtering for real compliance before measuring, and measuring distance along the corridor a vehicle actually drives, meant the gap segments identified in each state's analysis lined up with the same standard federal reviewers would apply, and running the same method across three states gave each DOT client a coverage picture that could be trusted on its own, not just visually plausible.